

Evaluating Expressions with RadicalsQuestion Bank 4.05 **ANSWER KEY**

*Show your work. Circle your final answer.***For questions 1 through 6, evaluate.**

1. $\sqrt{81}$

9

2. $\sqrt[3]{125}$

5

3. $\sqrt[3]{-8}$

-2*An odd root keeps the sign of its radicand.*

4. $\sqrt{\frac{4}{25}}$

 $\frac{2}{5}$ *The numerator and the denominator each get the root.*

5. $\sqrt{5^2 + 11}$

6*Finish under the bar: $25 + 11 = 36$.*

6. $\sqrt[3]{3^3 + 37}$

4 *$27 + 37 = 64$.*

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{100} \cdot 2^{-2}$

$$\frac{5}{2}$$

$$10 \cdot \frac{1}{4} \quad .$$

b. $\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$

$$\frac{17}{4}$$

$$\frac{1}{4} + 4 \quad .$$

c. $\sqrt[3]{-64} - 3^{-1}$

$$-\frac{13}{3}$$

$$-4 - \frac{1}{3} \quad .$$

8. Evaluate each expression.

a. $\sqrt[3]{27} - 1.4(\sqrt{3^2 - 5})$

$$0.2$$

$$3 - 1.4(2) \quad .$$

b. $\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$

$$-\frac{17}{9}$$

$$\frac{1}{9} - 2 \quad .$$

c. $\sqrt{4^3 - 39} \cdot 5^{-3}$

$$\frac{1}{25}$$

$$5 \cdot \frac{1}{125} \quad .$$

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 12$

$n = 144$

b. $\sqrt[3]{n} = -5$

$n = -125$

c. $\sqrt{n+7} = 9$

$n = 74$

$n + 7 = 81$.

Reading someone else's work.

10. Tonya writes $\sqrt{16+9} = \sqrt{16} + \sqrt{9} = 7$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that her method ignores.

$\sqrt{16+9} = \sqrt{25} = 5$. *The bar groups 16 + 9 into a single radicand, so the addition is finished before any root is taken.*

Her method would need the root of a sum to equal the sum of the roots, and it does not.

11. Which is the value of $\sqrt{36+64}$?

- ☐ A. 14
☐ B. 100
☒ C. 10
☐ D. 50

C

12. Select ALL of the expressions whose value is 3.

- ☐ A. $\sqrt{3^2+16}$
☒ B. $\sqrt[3]{2^2+23}$
☒ C. $\sqrt{4^2-7}$
☒ D. $\sqrt[3]{27}$
☐ E. $\sqrt[3]{-27}$
☒ F. $\sqrt{9}$

B, C, D and F

C's radicand is $16-7=9$ and B's is $4+23=27$.